

Course Code (Following BNQF): CSE 0533 1201 (for CSE) & PHY 0533 1201 (for EEE)

Course Title: Physics - II

Credit: 3.0

Rationale of the course: This course is to give students a broad understanding of physics and a foundation in problem solving and critical thinking. Topics covered in this course include Thermodynamics, Modern Physics and Optics.

Course Content:

Thermodynamics: Principle of temperature measurements: Platinum resistance thermometer, Thermo-electric thermometer, Pyrometer, Kinetic theory of gases: Maxwell's distribution of molecular speeds, Mean free path, Equipartition of energy, Brownian motion, van der Waal's equation of state, Review of the First law of thermodynamics and its application, Reversible & irreversible processes, Second law of thermodynamics, Carnot; Efficiency of heat engines, Carnot theorem, Entropy and Disorder, Thermodynamic Functions, Maxwell relations, Clausius-Clapeyron equation, Gibbs phase rule, Third law of thermodynamics.

Modern Physics: Interpretation of Bohr's postulates, Properties of nucleus, nuclear reactions, Fission. Fusion, Chain reaction, nuclear reactor, Wave particle duality, Compton effect, Photo-electric effect, Quantum theory of radiation, Radioactivity, Michelson-Morley's experiment, Frame of references, Galilean transformation, Lorentz transformations, Special theory of relativity & its consequences.

Light: Electromagnetic radiation, Pointing Vector, wave front, Huygens Principle, Interference of light, Young's double slit experiment, Displacements of fringes & its uses. Fresnel Bi-prism, Interference at wedge shaped films, Newton's rings, Interferometers, Diffraction of light: Fresnel and Fraunhofer diffraction. Diffraction by single slit. Diffraction from a circular aperture, Resolving power of optical instruments, Diffraction at double slit & N-slits-diffraction grating. Polarization: Production & analysis of polarized light, Brewster's law, Malus law, Polarization by double refraction. Retardation plates, Nicol prism. Optical activity. Polarimeters, Polaroid.

Course Learning Outcomes (CLOs):

CLOs	CLOs Title
CLO 1	Specify the basic terms related to thermodynamics, modern physics and light.
CLO 2	Identify and formulate the various quantities to explain physical and engineering phenomena.
CLO 3	Apply critical thinking and problem-solving skills to solve a variety of problems concerning thermodynamics, modern physics and light.
CLO 4	Demonstrate analytical proficiency to establish the physical and mathematical logic behind natural phenomena observed in everyday life.

Mapping CLOs with PLOs and Domains-Taxonomy Levels:

CLOs	Corresponding PLOs	Domains and Taxonomy Levels	
		Blooms	BNQF
CLO 1	PLO 1	C1	F1
CLO 2	PLO 1	C2	F2
CLO 3	PLO 2	C3	F3
CLO 4	PLO 4	C4	F6

Course Plan specifying content, CLOs, Co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week	Topic	Teaching Strategy	Assessment Technique	Alignment with CLOs
01	Principle of temperature measurements: Platinum resistance thermometer, Thermo-electric thermometer, Pyrometer	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Assignment Mid-term Exam	1 & 2
02	Kinetic theory of gases: Maxwell's distribution of molecular speeds, Mean free path, Equipartition of energy, Brownian motion, van der Waal's equation of state	Interactive discussion Group activity Tutorials Problem Solving	Oral Test Home works Assignment Mid-term Exam	2 & 3
03	Review of the First law of thermodynamics and its application	Interactive discussion Group Presentation Tutorials	Class Test Home works Mid-term Exam	1
04	Reversible & irreversible processes, Second law of thermodynamics, Carnot; Efficiency of heat engines, Carnot theorem, Entropy and Disorder	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Mid-term Exam	2
05	Thermodynamic Functions, Maxwell relations, Clausius-Clapeyron equation, Gibbs phase rule, Third law of thermodynamics	Interactive discussion Demonstration Group activity Tutorials	Oral Test Home works Assignment Mid-term Exam	3 & 4
06	Interpretation of Bohr's postulates, Properties of nucleus, nuclear reactions, Fission. Fusion, Chain reaction, nuclear reactor	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Mid-term Exam	1
07	Wave particle duality, Compton effect, Photo-electric effect, Quantum theory of radiation	Interactive discussion Demonstration Group activity Tutorials	Class Test Home works Mid-term Exam	2
08	Radioactive disintegration	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Assignment Final Exam	1 & 3
09	Michelson-Morley's experiment, Frame of references, Galilean transformation, Lorentz transformations	Interactive discussion Group activity Tutorials Problem Solving	Oral Test Home works Assignment Final Exam	3 & 4
10	Special theory of relativity & its consequences	Interactive discussion Group Presentation Tutorials	Class Test Home works Final Exam	3 & 4

11	Interference of light, Young's double slit experiment, Displacements of fringes & its uses. Newton's rings, Thin film	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Final Exam	1 & 2
12	Diffraction of light: Fresnel and Fraunhofer diffraction. Diffraction by single slit, Resolving power, Diffraction grating.	Interactive discussion Demonstration Group activity Tutorials	Oral Test Home works Assignment Final Exam	2 & 3
13	Polarization: Production & analysis of polarized light, Brewster's law, Malus law, Polarization by double refraction.	Interactive discussion Group activity Tutorials Problem Solving	Oral Test Class works Final Exam	2
14	Nicol prism, Mathematical Problems	Interactive discussion Group Presentation Tutorials	Class Test Home works Final Exam	3 & 4
15	Review Class/Problem Solving	Interactive discussion Group Presentation Tutorials	Oral Test Class works Final Exam	2 & 4

Ref. Book:

- Physics for Engineers (Part – 1 & Part – 2) – Gias Uddin Ahmed
- Halliday Resnick - Fundamental of Physics, 2nd edition